

CDS505

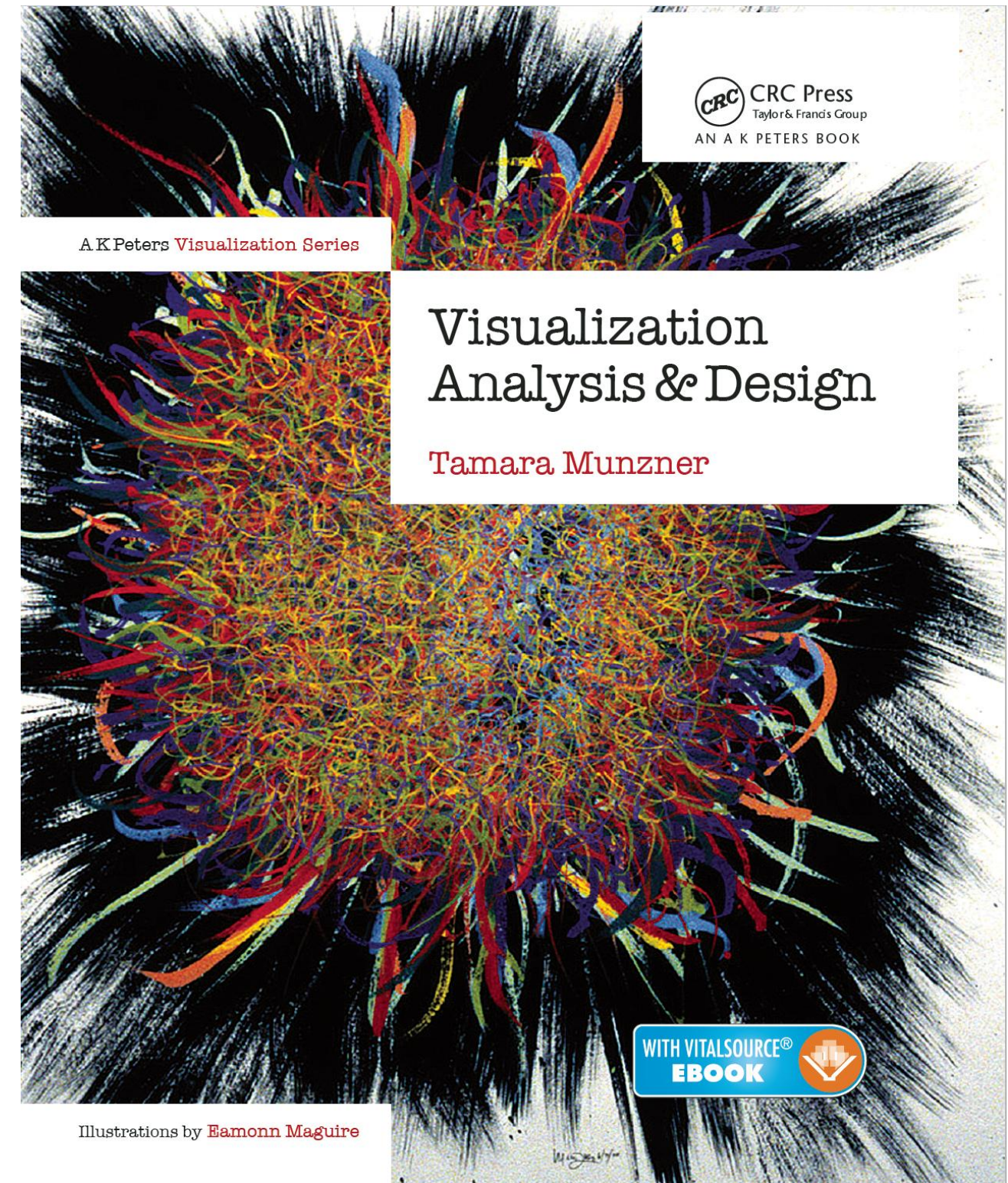
Data Visualization & Visual Analytics

Task Abstraction

*Nur Intan Raihana Ruhaiyem, PhD.
School of Computer Sciences
Universiti Sains Malaysia*

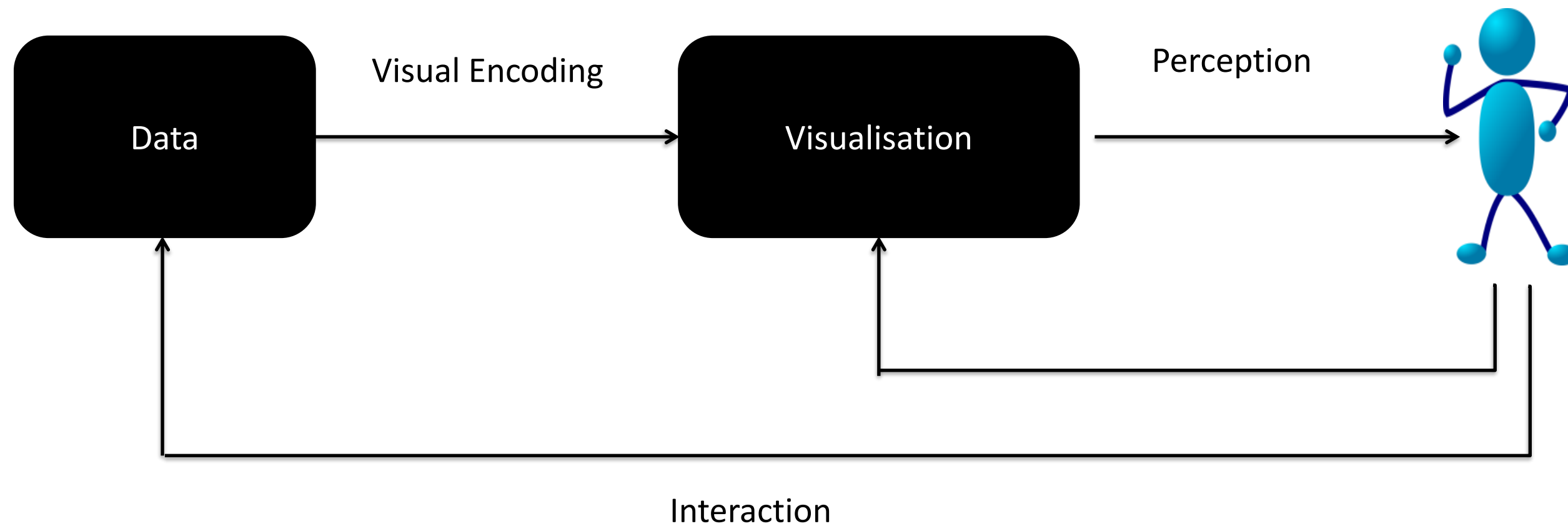
Website:

<https://cs.usm.my/index.php/faculty-member/194-nur-intan-raihana-ruhaiyem-dr>



- ✓ Recap previous class
- ✓ Task Abstraction
- ✓ Class Activity

Data Visualisation Pipeline



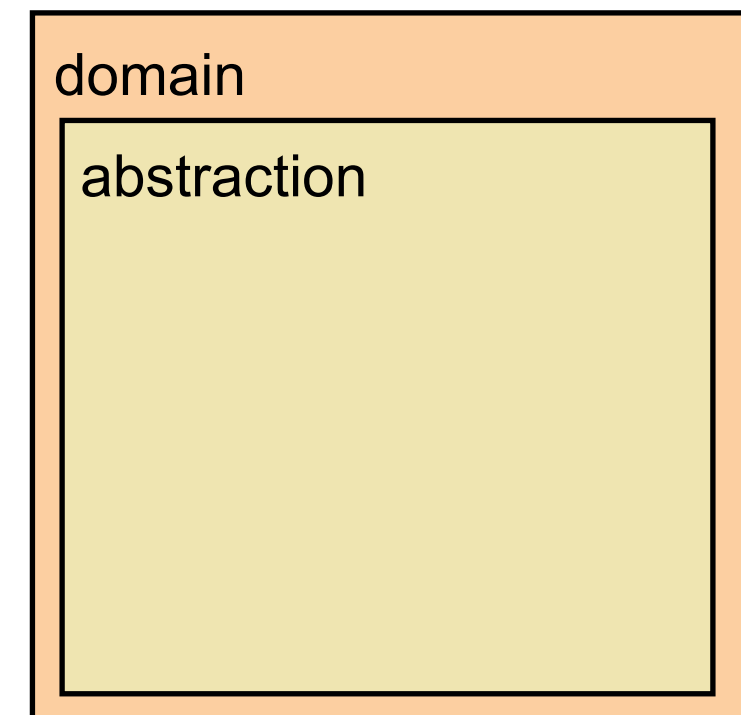
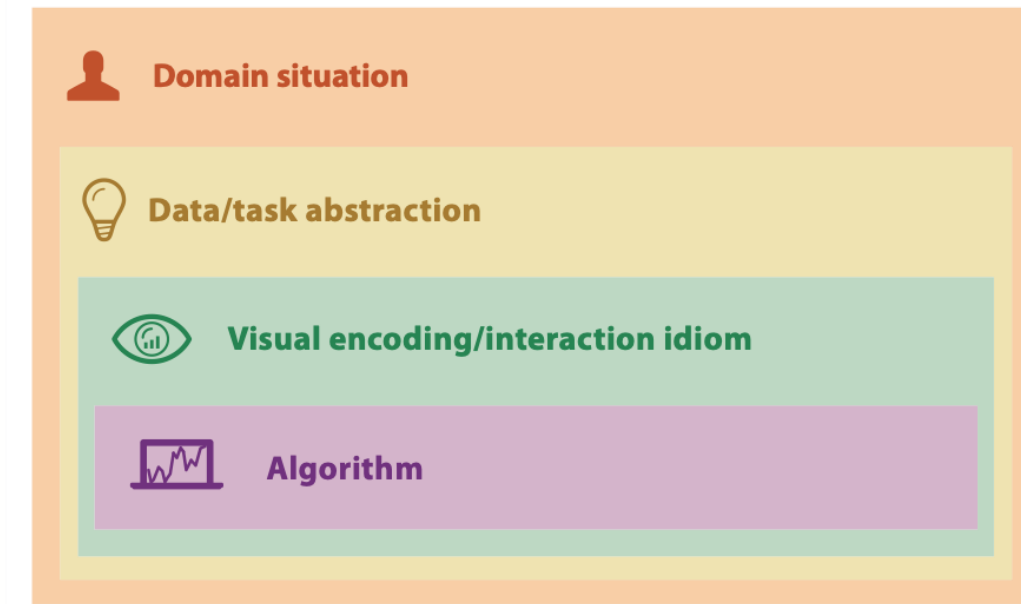
- **Explanatory**
 - To tell story about known facts; to better explain something to someone or an audience i.e. to communicate
- **Exploratory**
 - Some aspects of the phenomena is not well known; hence the purpose here is extracting information/knowledge from data to help person answer some questions about the phenomena
- **Confirmatory**
 - Similar to above BUT with the purpose to verify or validate finding about the phenomena

Computer-based visualization systems provide visual representations of datasets designed to help people carry out tasks more effectively.

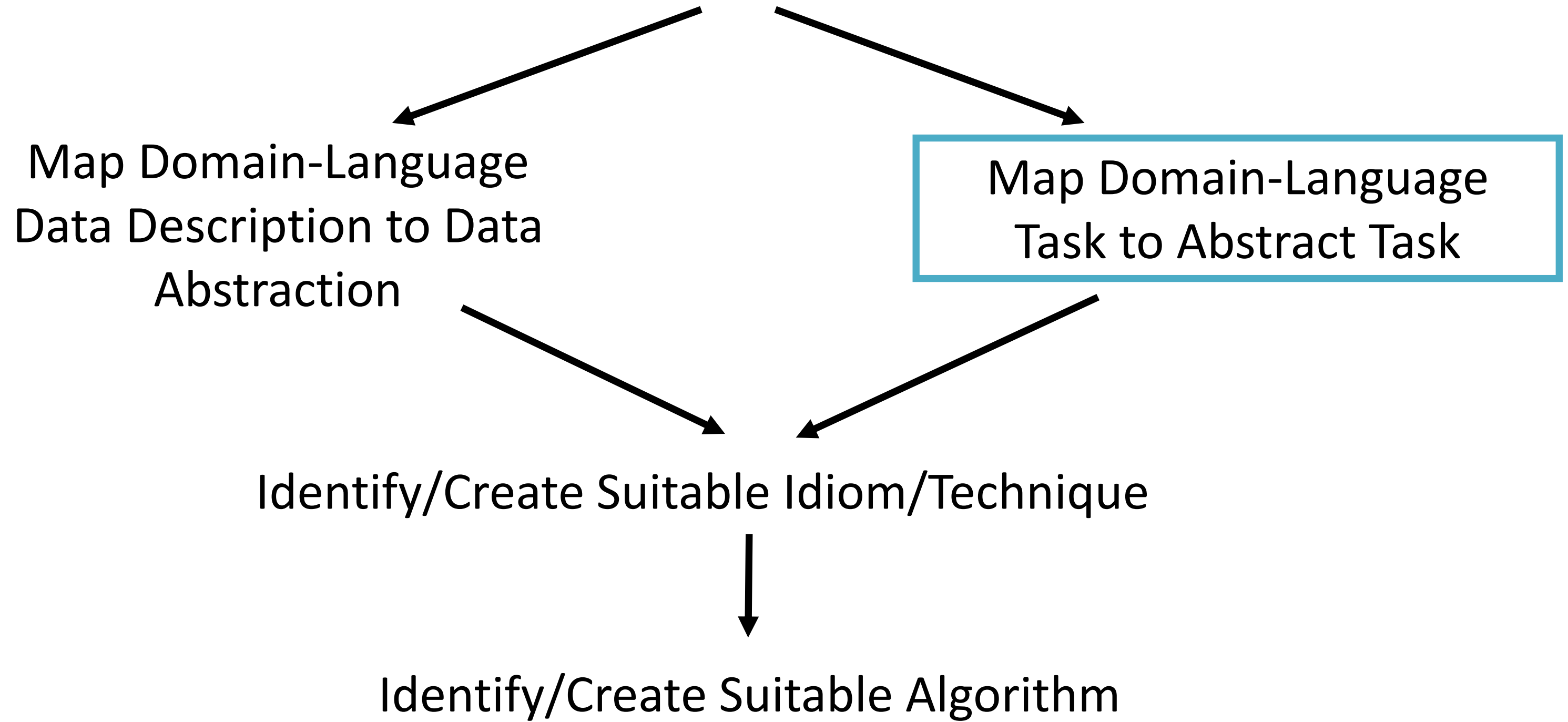
- How did we **decide** to use those particular visual channels?
- How **many** more visual channels are there?
- What kinds of **information** and how much information can each **channel** encode?
- Are some channels **better** than others?
- Can all of the channels be used **independently** or do they **interfere** with each other?

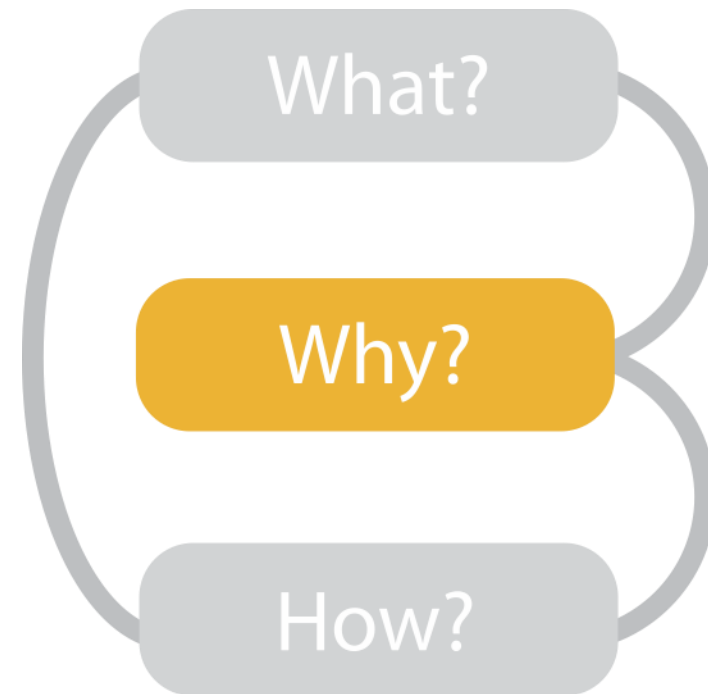
From Domain to Abstraction

- **Domain characterization:** details of application domain
 - group of users, target domain, their questions & data
 - varies wildly by domain
 - must be specific enough to get traction
 - domain questions/problems
 - break down into simpler abstract tasks
- **Abstraction:** data & task
 - map *what* and *why* into generalized terms
 - identify tasks that users wish to perform, or already do
 - find data types that will support those tasks
 - possibly transform /derive if need be



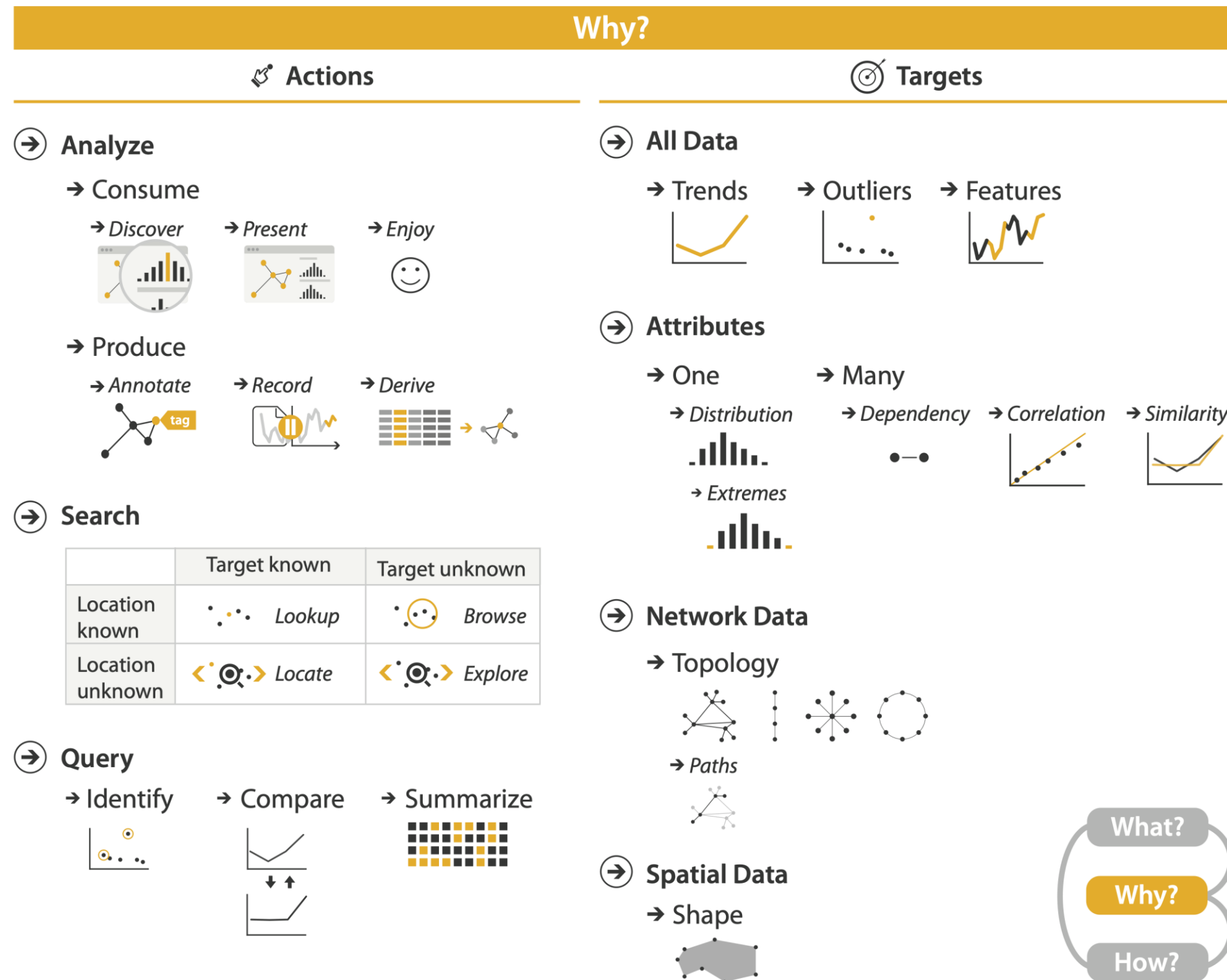
Design Process





- This is about the usage of vis – the **why** part?
- Why Abstraction?
 - Hide details
 - Genericity (non-specificity)
- Two usage terms:
 - Actions
 - Targets

The Big Picture – Task Abstraction



{action, target} pairs

- *discover distribution*
- *compare trends*
- *locate outliers*
- *browse topology*

- **Very high-level pattern**
- **Actions**
 - analyze
 - high-level choices
 - search
 - find a known/unknown item
 - query
 - find out about characteristics of item
- **Targets**
 - what is being acted on

{action, target} pairs

- *discover distribution*
- *compare trends*
- *locate outliers*
- *browse topology*

Actions: Analyse

- **Consume**

- discover vs present
 - classic split
 - aka explore vs explain
- enjoy
 - newcomer
 - aka casual, social

- **Produce**

- annotate, record
- derive
 - crucial design choice

→ Analyse

→ Consume

→ *Discover*



→ *Present*

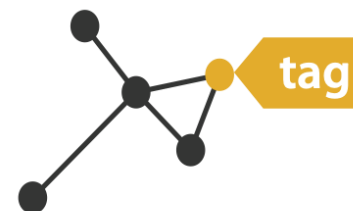


→ *Enjoy*

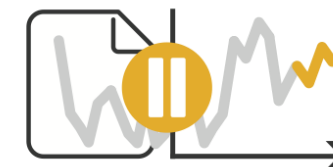


→ Produce

→ *Annotate*



→ *Record*



→ *Derive*



Actions: Analyse

- **Consume**
 - discover vs present
 - classic split
 - aka explore vs explain
 - enjoy
 - newcomer
 - aka casual, social

- **Produce**
 - annotate, record
 - derive
 - crucial design choice

➔ Analyse

➔ Consume

➔ *Discover*



➔ *Present*

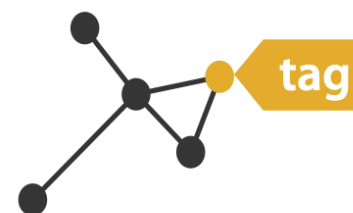


➔ *Enjoy*



➔ Produce

➔ *Annotate*



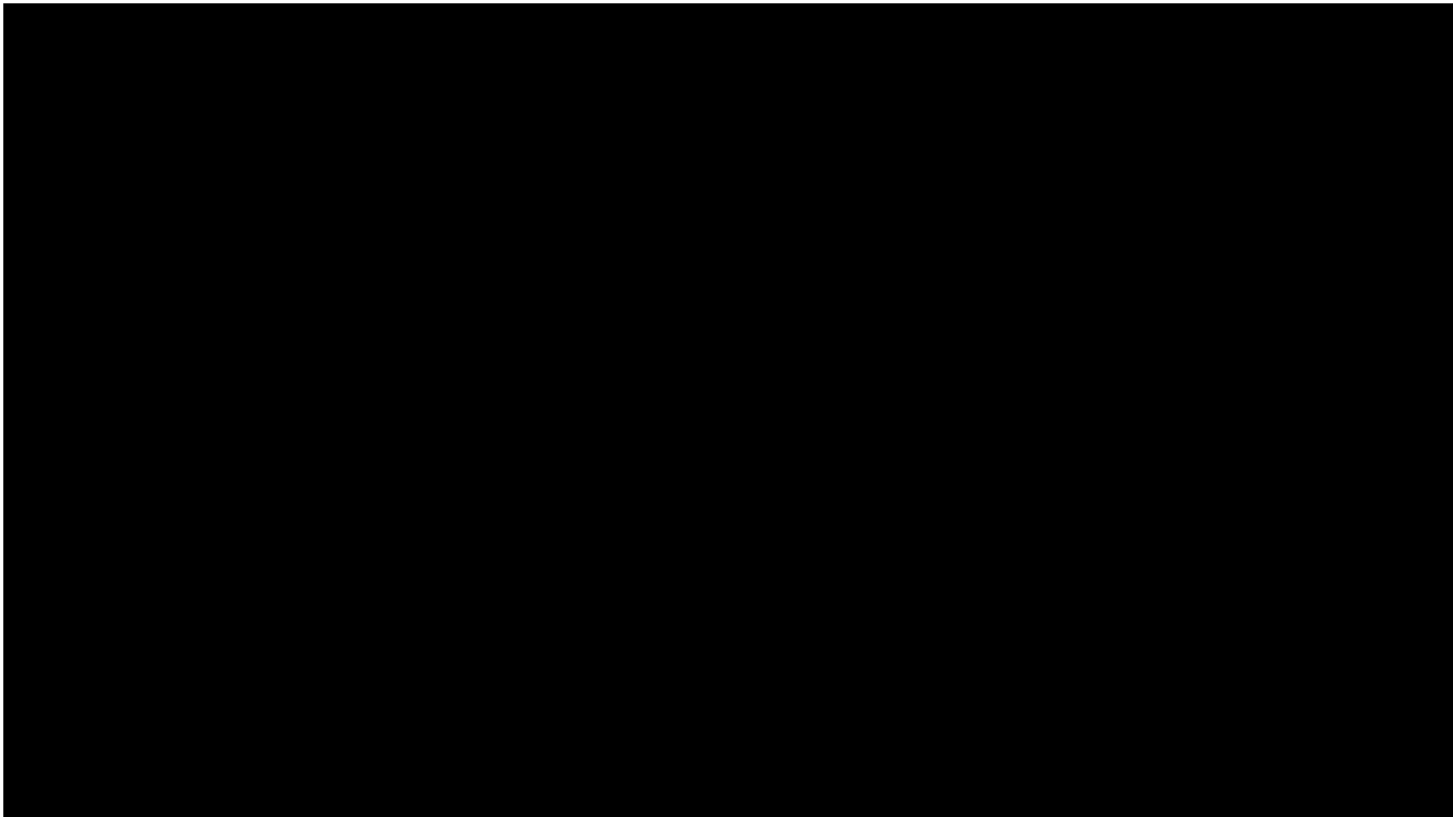
➔ *Record*



➔ *Derive*



Tableau Prep: Example of Derive

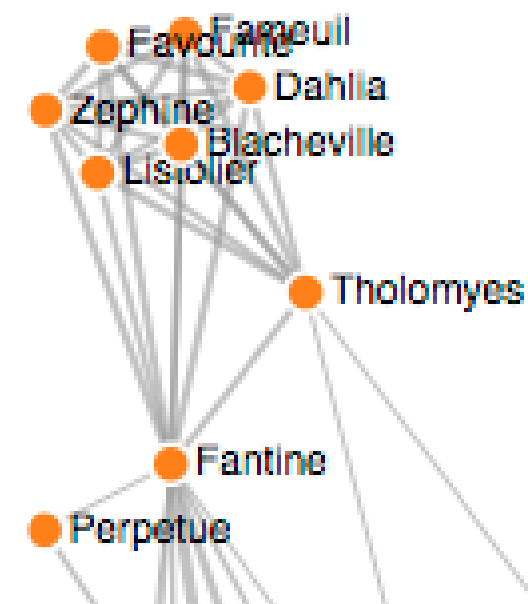


<https://www.youtube.com/watch?v=8hHf32pMeu8>

- **What does user know?**
 - target, location
- **Lookup**
 - ex: word in dictionary
 - alphabetical order
- **Locate**
 - ex: keys in your house
 - ex: node in network
- **Browse**
 - ex: books in bookstore
- **Explore**
 - ex: find cool neighborhood in new city

➔ **Search** - to search for elements of interest within the vis idiom

	Target known	Target unknown
Location known	 <i>Lookup</i>	 <i>Browse</i>
Location unknown	 <i>Locate</i>	 <i>Explore</i>

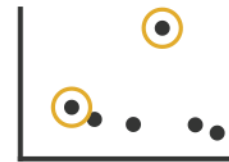


<https://bl.ocks.org/heybignick/3faf257bbbbc7743bb72310d03b86ee8>

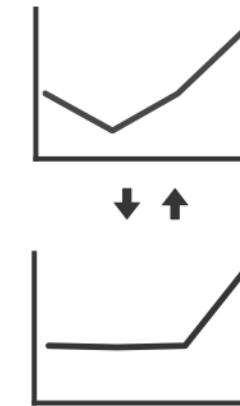
- How much of the data matters?
 - **One**: identify
 - **Some**: compare
 - **All**: summarize

➔ Query

➔ Identify



➔ Compare



➔ Summarize



- Independent choices for each of these three levels
 - analyze, search, query
 - mix and match

Actions

➔ Analyze

➔ Consume

➔ Discover



➔ Present



➔ Enjoy



➔ Produce

➔ Annotate



➔ Record



➔ Derive



➔ Search

	Target known	Target unknown
Location known	 Lookup	 Browse
Location unknown	 Locate	 Explore

➔ Query

➔ Identify



➔ Compare



➔ Summarize



Task abstraction: Targets

→ All Data

→ Trends



→ Outliers



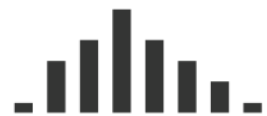
→ Features



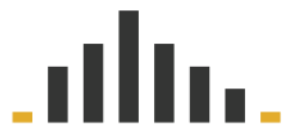
→ Attributes

→ One

→ *Distribution*



→ *Extremes*

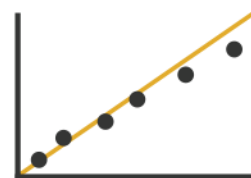


→ Many

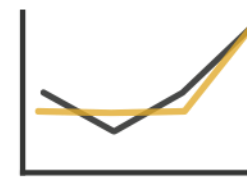
→ *Dependency*



→ *Correlation*

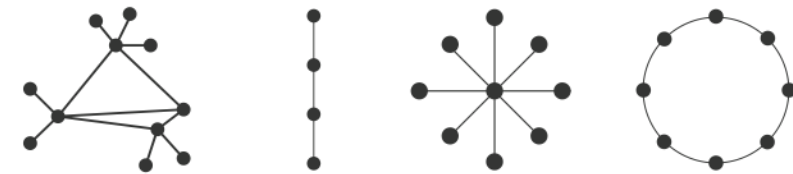


→ *Similarity*



→ Network Data

→ Topology



→ *Paths*



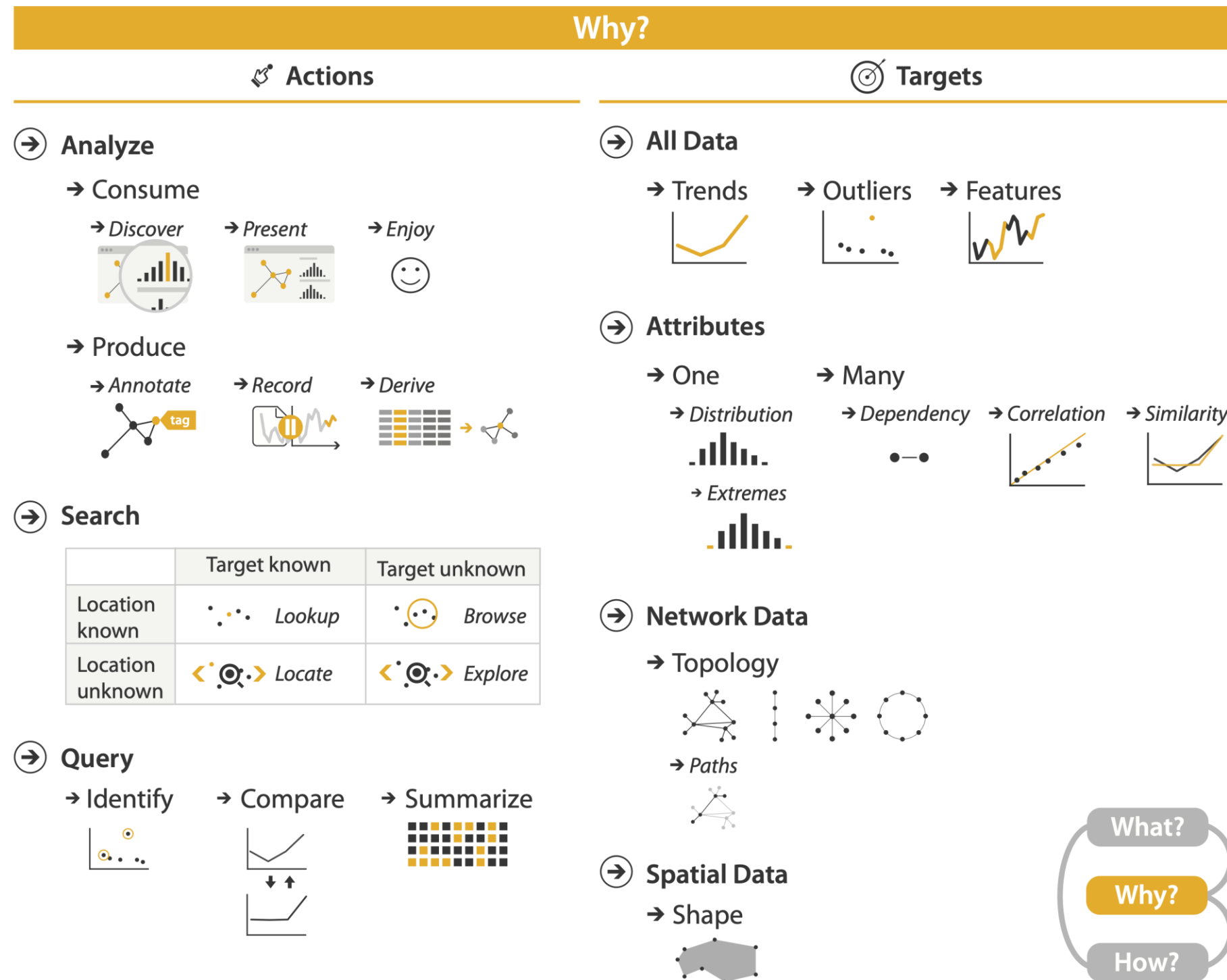
→ Spatial Data

→ Shape



- These {action, target} pairs are good starting point for vocabulary
 - but sometimes you'll need more precision!
- Rule of thumb
 - systematically remove all domain jargon
- Interplay: task and data abstraction
 - need to use data abstraction within task abstraction
 - to specify your targets!
 - but task abstraction can lead you to transform the data
 - iterate back and forth
 - first pass data, first pass task, second pass data, ...

The Big Picture – Task Abstraction



{action, target} pairs

- *discover distribution*
- *compare trends*
- *locate outliers*
- *browse topology*

Specificity vs Genericity

Who has a goal or makes a design choice : the designer of the vis or the end user of the vis?

- Your neighbour is your partner, choose one role each.
- Think individually to justify your role, list reasons to support your role in design choice.
- Take turn to explain your role crucial points in the design choice
- Explain to the class your points either individually or consensus as a pair